

IIT BOMBAY
CHEMICAL ENGINEERING

CL420: Introduction to Biochemical Engineering

Prof. Pramod P Wangikar

Classes on Mon, Tues, Thurs

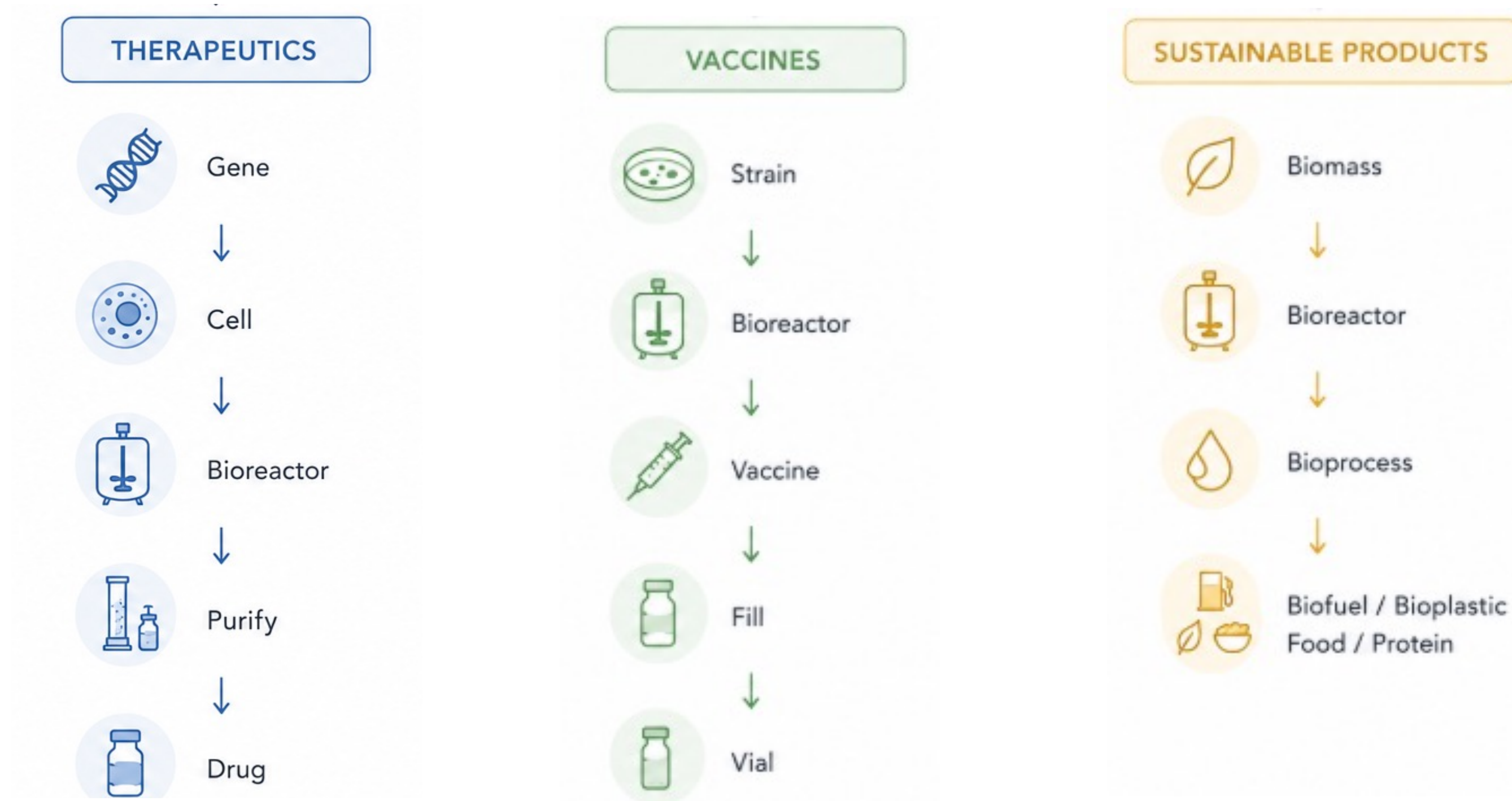
10:35 AM to 11:30 AM

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Credits for PPT: Claude and TAs

What does a biochemical engineer do?

The same pipeline as chemical engineering — grow, make, recover, purify
But by using cells for production of wildly different products

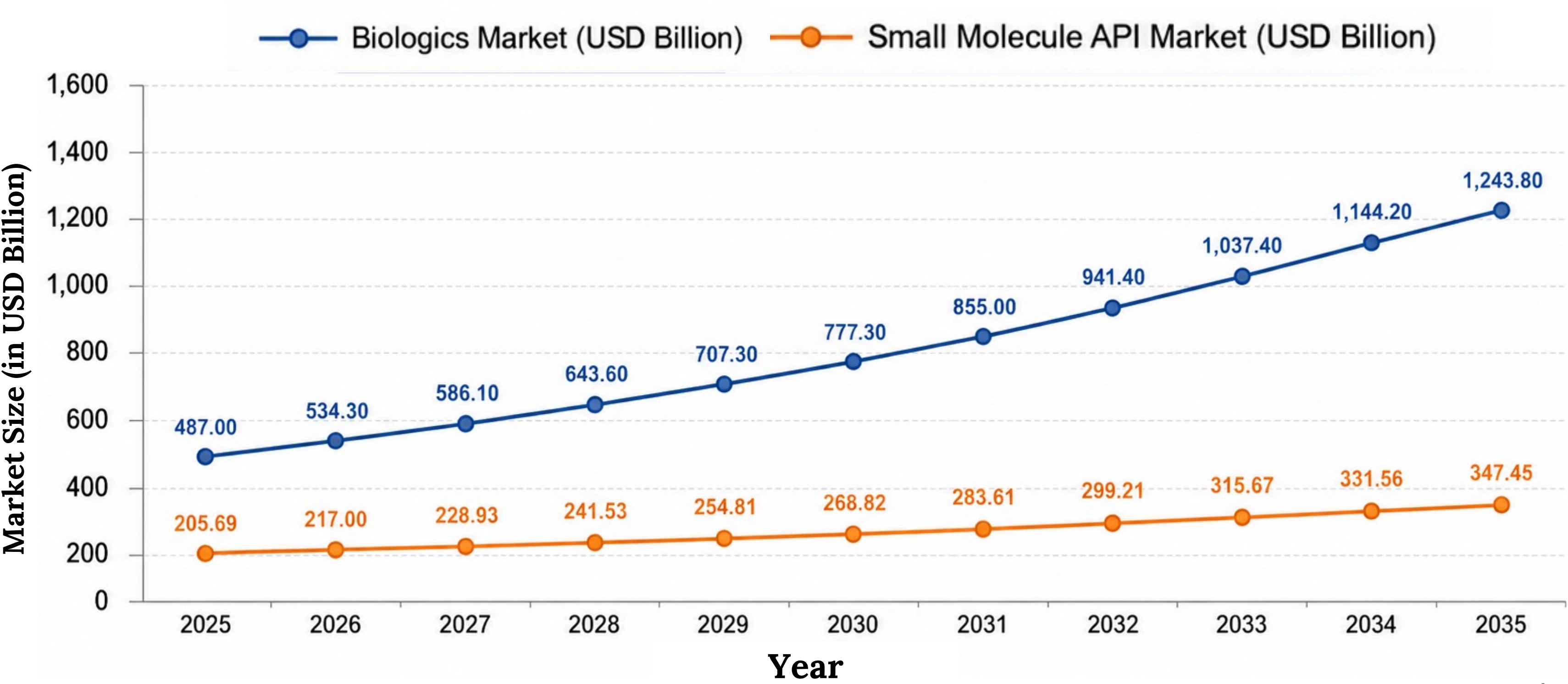


Whatever the product, the engineering questions are the same: how big, how fast, how much, how pure.

WHERE THE INDUSTRY IS GOING

Market growth

Biologics are the fastest-growing, most engineer-intensive segment of pharma.



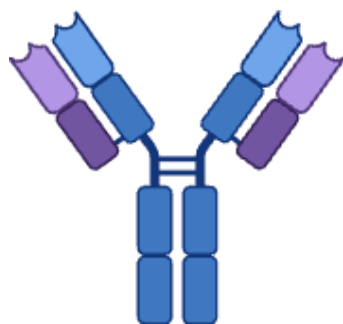
Blockbuster biologics

PRODUCT	TYPE	INDICATION	2025 GLOBAL SALES
Keytruda	Monoclonal Antibodies	Cancer	\$31.7 Billion
Mounjaro	Peptide	Diabetes	\$32.3 Billion
Ozempic	Peptide	Diabetes	\$19.6 Billion
Wegovy	Peptide	Obesity	\$12.5 Billion
Dupixent	Monoclonal antibody	Asthama, COPD	\$18.1 Billion
Casgevy	CRISPR	Sickle cell disease	\$116 Million
Luxturna	Gene therapy	Inherited blindness	\$45 Million

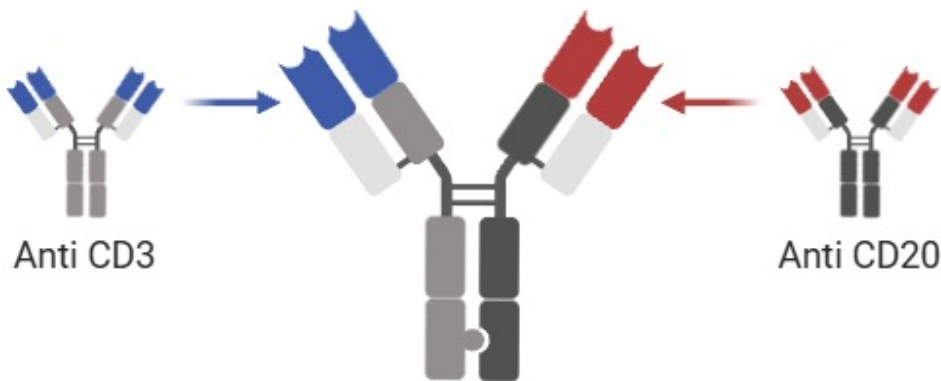
Emerging therapeutic modalities

Every product needs a different manufacturing process — that's where engineers come in.

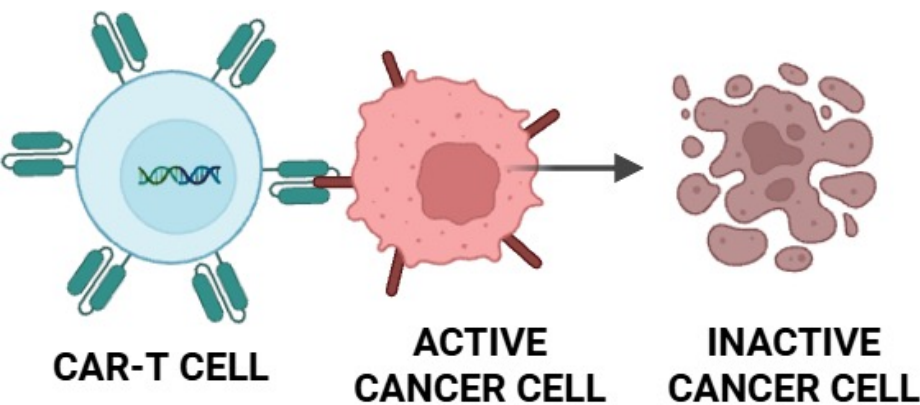
Monoclonal antibodies



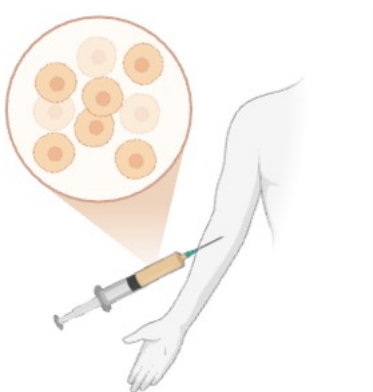
Bispecifics



CAR-T



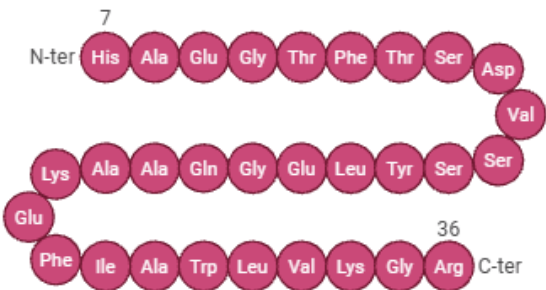
Stem-cell therapies



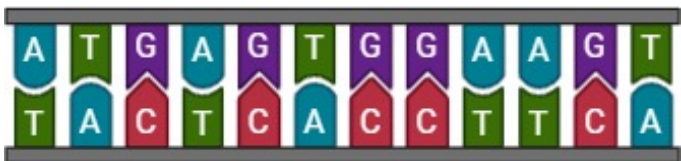
CRISPR



Peptides



Oligonucleotides



INDIA'S BIOMANUFACTURING ECOSYSTEM

India in the picture

PUNE

Serum Institute

World's largest vaccine maker by volume.

BENGALURU

Biocon

Global insulin & biosimilars — world's first interchangeable biosimilar insulin glargine

PUNE

Gennova

GEMCOVAC — India's first mRNA vaccine.

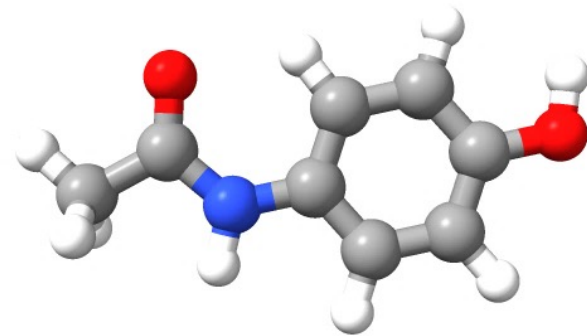
HYDERABAD

Dr. Reddy's

Biosimilars including rituximab and trastuzumab.

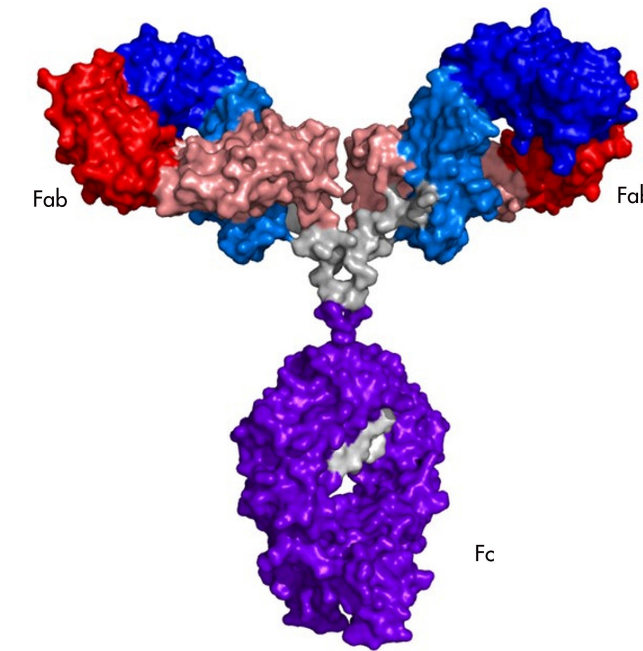
But these biologically-made products are hard to produce

Small molecule: Paracetamol



- Size ~180 Da
- Chemical synthesis
- Simple, well-defined
- Easy to characterize

Biologics: IgG Antibody



- ~150 KDa
- Made by living cells
- Complex folding & modifications are difficult
 - Hard to characterize & purify

WHY THIS FIELD MATTERS

Why does making a protein cost so much more than making a pill?

BIOLOGICS FACILITY

\$200–500M

Living cells as the production line

SMALL-MOLECULE FACILITY

\$30–100M

Fixed chemistry, standardized unit operations.

THE INDIAN SCALE: Serum Institute, Pune

1.5B+ doses/year — world's largest vaccine maker by volume

~65% of the world's children get ≥ 1 SII vaccine, across 170+ countries

4B annual dose capacity at Manjri; Covishield hit 100M doses/month at COVID peak

What is biochemical engineering?

A bridge discipline — the quantitative tools of chemical engineering applied to living cells.

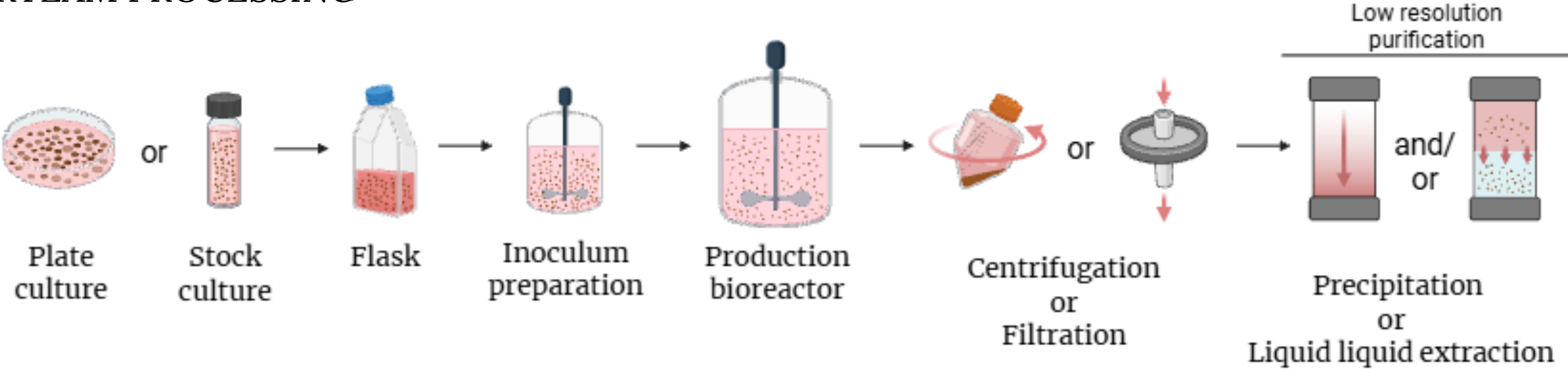


Reaction Engineering	rate laws & reactor design, now for cells
Transport Phenomena	oxygen transfer, mixing, heat removal
Thermodynamics	yields, energetics, feasibility
Process Control	holding pH, DO, temperature on target
Molecular & Systems Biology	what the cell is doing inside
Data Science	turning sensor streams into decisions

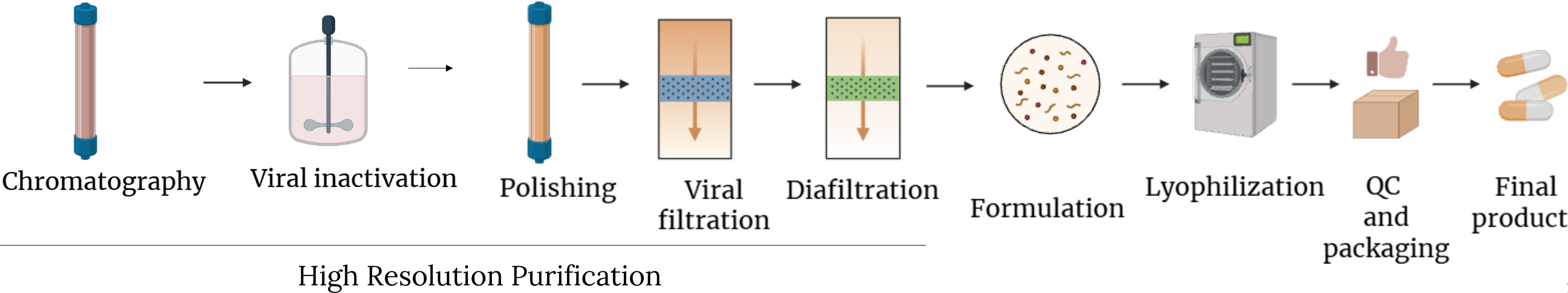
You already know most of these tools — this course applies them to a living, variable cell.

The manufacturing pipeline

UPSRTEAM PROCESSING



DOWNSRTEAM PROCESSING



CHOOSING THE HOST

Cell factories

E. coli

- Fast, cheap
no complex
modifications.

Yeast

- Robust, secretes
protein
simple
glycosylation.

CHO Cells

- Industry workhorse
for antibodies
human-like
glycosylation.

HEK293

- Human line;
complex proteins.

Insect cells

- High-yield for
some vaccines.

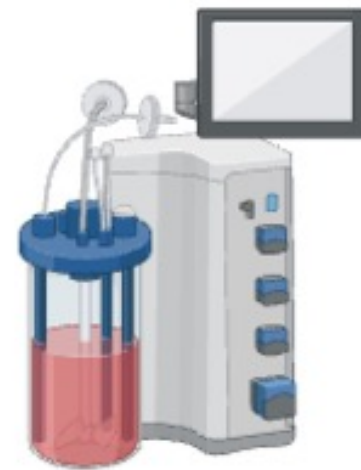
Choosing the host is itself an engineering decision.

UPSTREAM

Upstream processing



Media
shake flask



Inoculum
bench scale



Seed train
pilot scale

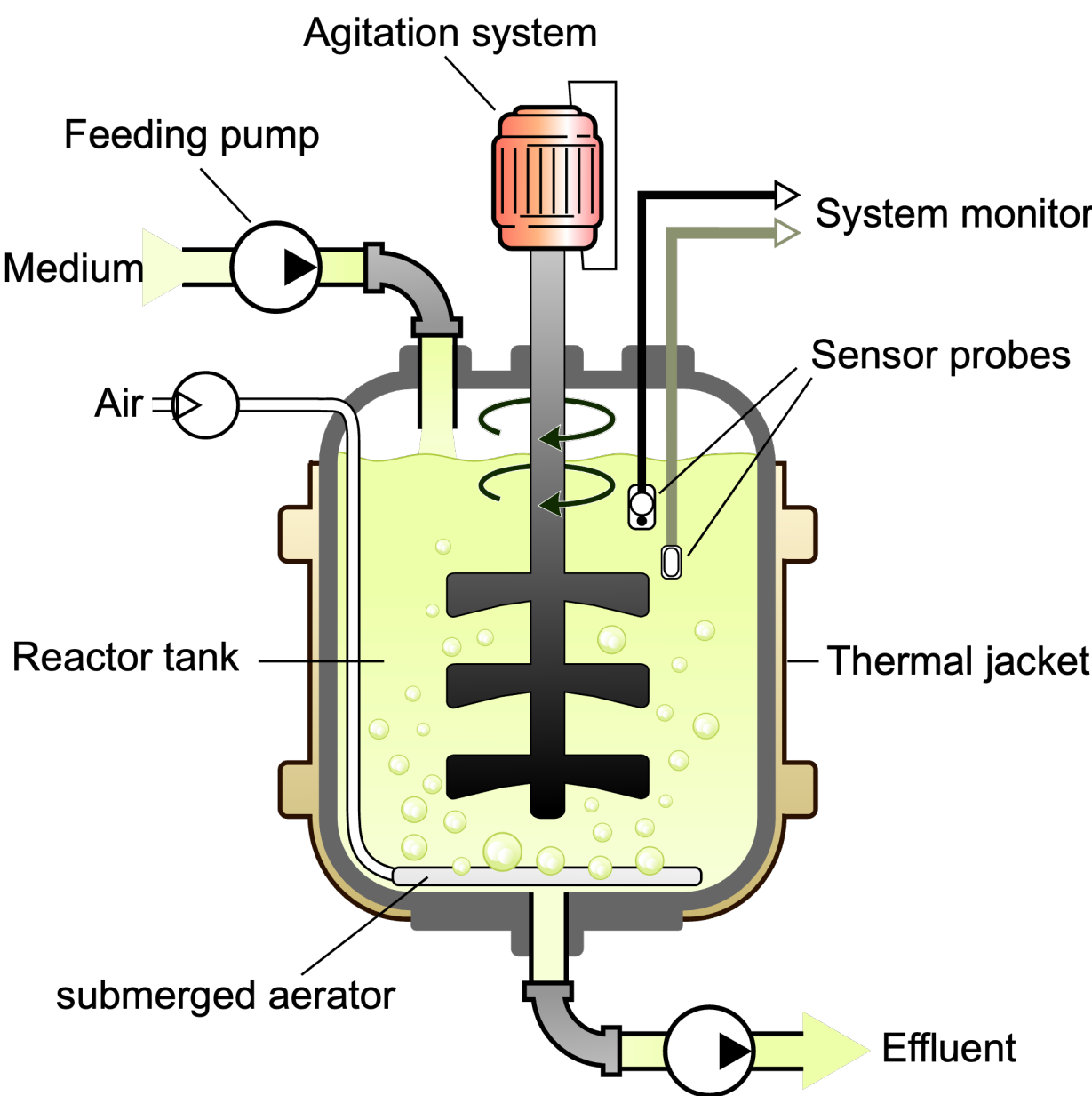


Production bioreactor
production scale

Scale-up proceeds shake flask → bench → pilot → production, each roughly 10× larger than the last.

THE HEART OF UPSTREAM

A bioreactor



Source: Wikipedia

Impeller	Mixing
Sparger	Oxygen delivery
Sensors	pH, DO, temperature
Cooling jacket	Heat removal
Feed port	Nutrient addition

Oxygen transfer

Mixing

Sterility

We'll calculate all three later in the course.

THE HEART OF UPSTREAM

A bioreactor



Source: Stock exchange

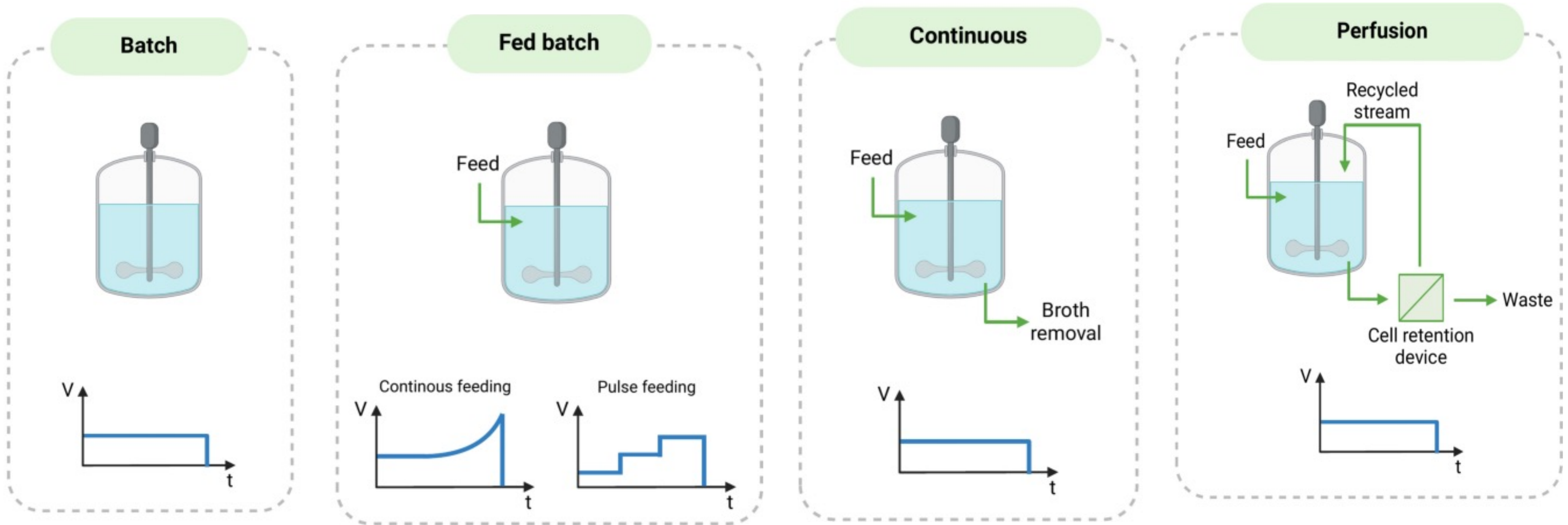


Source: Cultipliy

mAb treatments against rheumatoid arthritis

- ❑ adalimumab (Humira),
- ❑ infliximab (Remicade), and
- ❑ tocilizumab (Actemra)

Modes of operation



All nutrients added at the start.

Nutrients fed over time – the industry default for mAbs.

Fresh media in, product out – steady state.

Mode choice affects titre, quality and cost.

SEPARATING CELLS FROM PRODUCT

Harvest

BY DENSITY

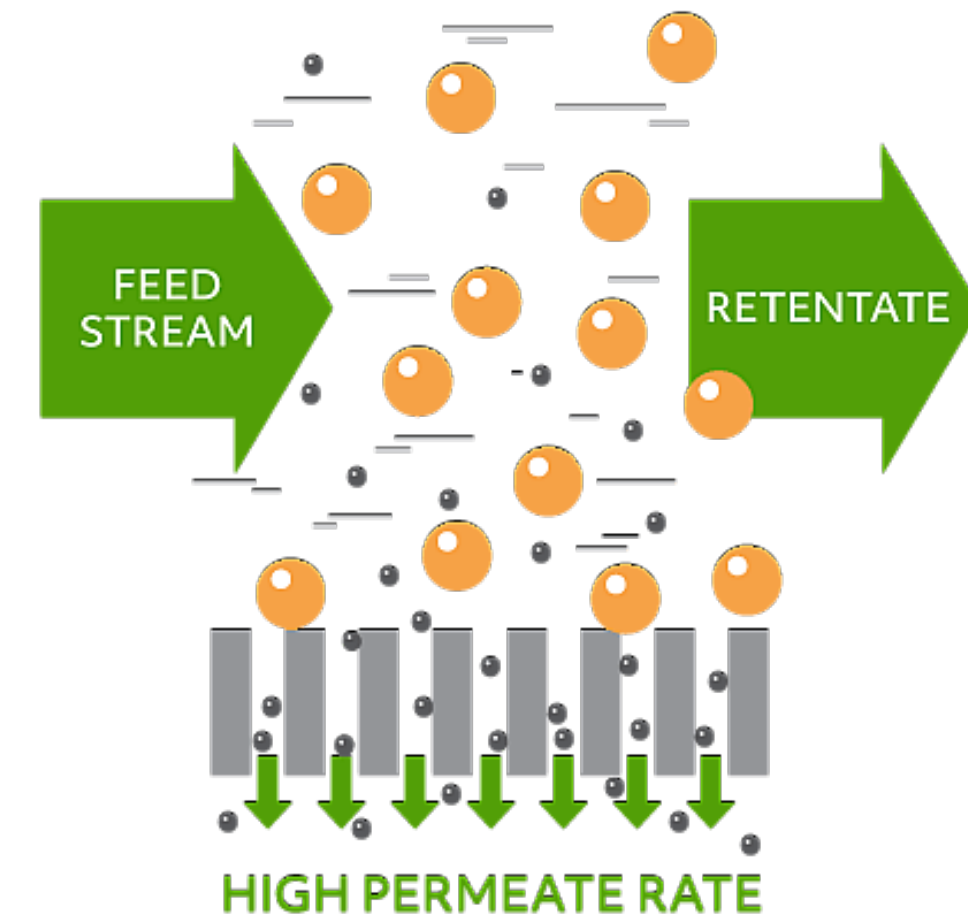
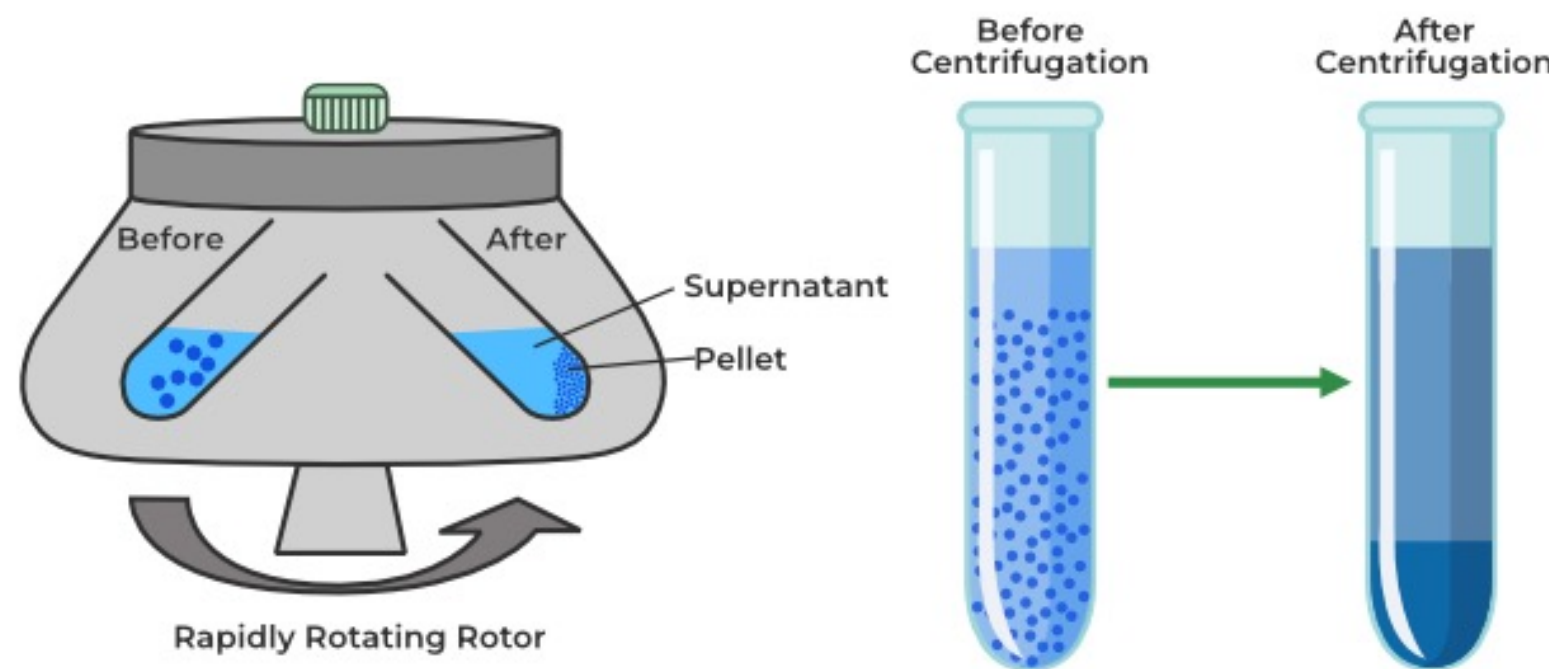
Centrifugation

Spins cells out of suspension using density differences.

BY SIZE

+ Depth / tangential-flow filtration

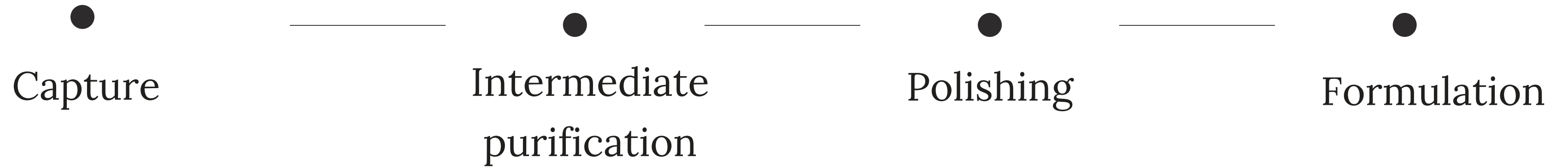
Filters cells and debris out based on particle size.



The bridge from upstream to downstream.

DOWNSTREAM

Downstream processing



Each step trades yield for purity.

Chromatography

AFFINITY

Protein A

Captures antibodies specifically.

BY CHARGE

Ion exchange

Separates on net surface charge.

BY HYDROPHOBICITY

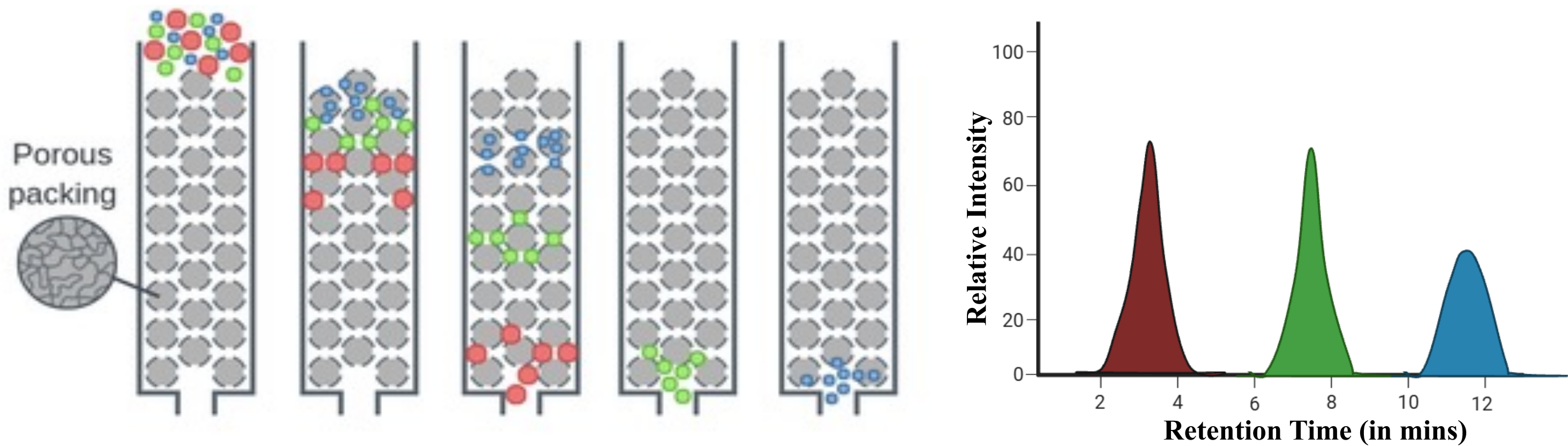
Hydrophobic Interaction

Separates on surface hydrophobicity.

BY SIZE

Size exclusion

Separates on molecular size.

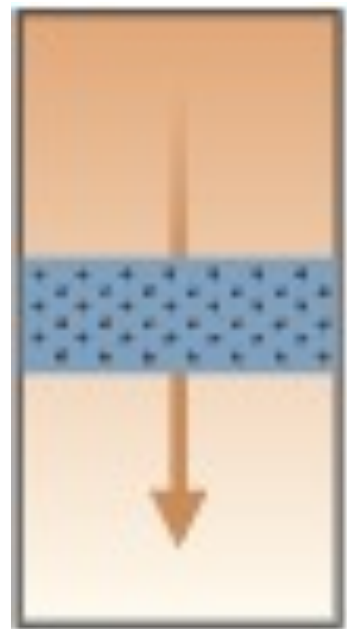


Schematic for Size Exclusion Chromatography

Virus removal & fill-finish

Nanofiltration

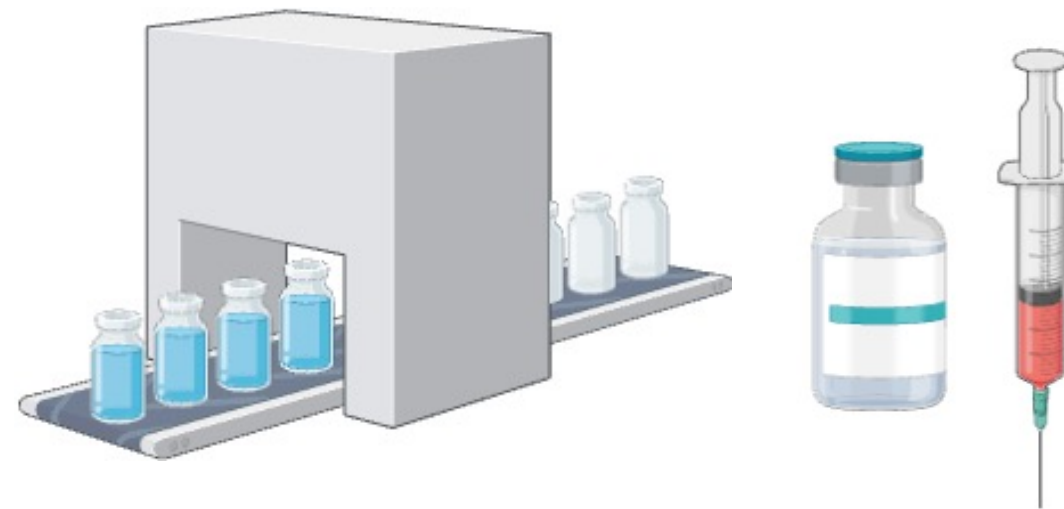
Removes viral contaminants for safety.



Viral Filtration

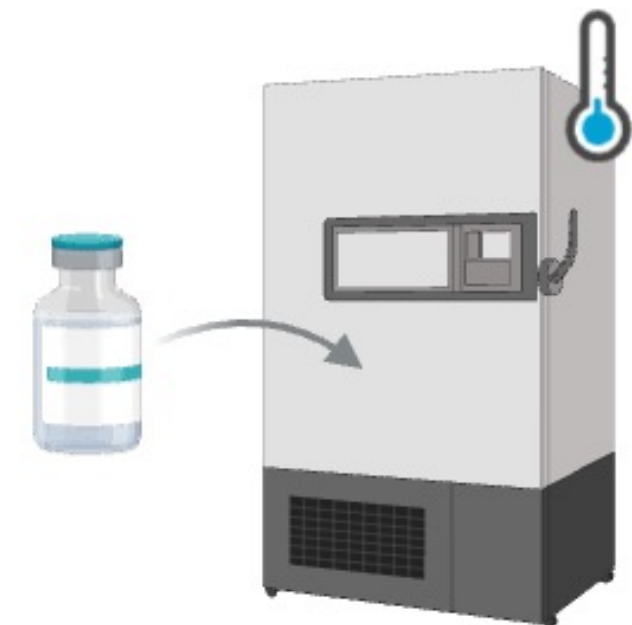
Sterile fill

Vials or prefilled syringes under aseptic conditions.



Cold chain

Continuous refrigerated storage and transport.



Cold chain is a real engineering & logistics constraint — it decides which markets a drug can reach.

Where does the money go?



● Downstream	45%
● Upstream	25%
● Quality control	20%
● Facilities	10%

*Downstream dominates cost even though this course focuses upstream —
hold that tension in mind.*

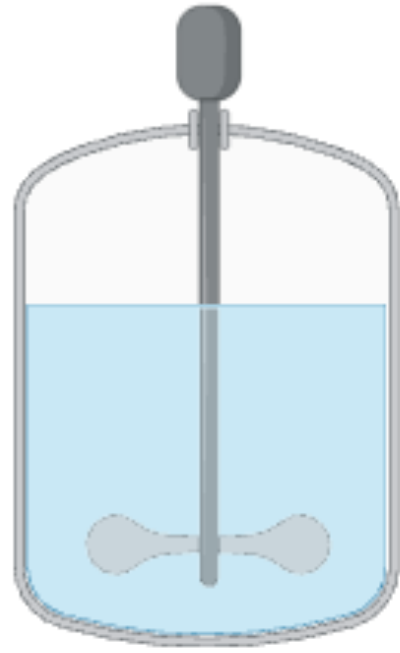


If you had to cut manufacturing cost by 20%,
which stage would you attack first — and why?

Think-pair-share, 2 min. (*5-min break after.*)

A DIFFERENT KIND OF REACTOR

Cells are not chemical reactors



A CSTR

- Fixed kinetics
- Well-defined rate laws
- Reproducible



A living cell population

- A living catalyst
- Batch-to-batch variability
- Responds to its environment

No two cell populations are identical — a real departure from the reactors you modelled in CRE.

WHAT WE ACTUALLY ASK

The typical engineering questions

-
- How much oxygen does this bioreactor need?
-
- How fast should we agitate?
-
- How much glucose feed, and when?
-

Same “how much / how fast / how big” as any process course.

QUALITY, PREVIEWED

Critical process parameters & quality attributes

CPP — WHAT WE CONTROL

Temperature

pH

Dissolved oxygen

Feed rate

Agitation



CQA — WHAT RESULTS

Aggregation

Glycosylation

Purity

Potency

A small CPP shift can move a CQA and fail a whole batch — we build the maths for this later.

“You cannot optimize what you cannot measure.”

Process Analytical Technology (PAT) — online, at-line and offline sensors turn a living process into a measurable one.

Measurement is the thread that ties mass transfer, kinetics, control and metabolomics together.

THE MODERN TOOLKIT

Digital bioprocessing & metabolomics

Digital twins

A live model runs alongside the real reactor.

ML soft-sensors

Predict hard-to-measure states from easy signals.

Media & metabolite analysis

Metabolomics reveals what the cell is actually doing.

Feed optimisation

Data-driven feeding strategies lift titre.

MEASUREMENT PAYS OFF

Case study: CHO process optimization

BEFORE

[X g/L]

baseline titre



AFTER

[Y g/L]

with better monitoring & control

WHAT YOU'LL BE ABLE TO DO

The questions this course answers

REACTOR SIZING

How big?

KINETICS

How fast?

STOICHIOMETRY & YIELD

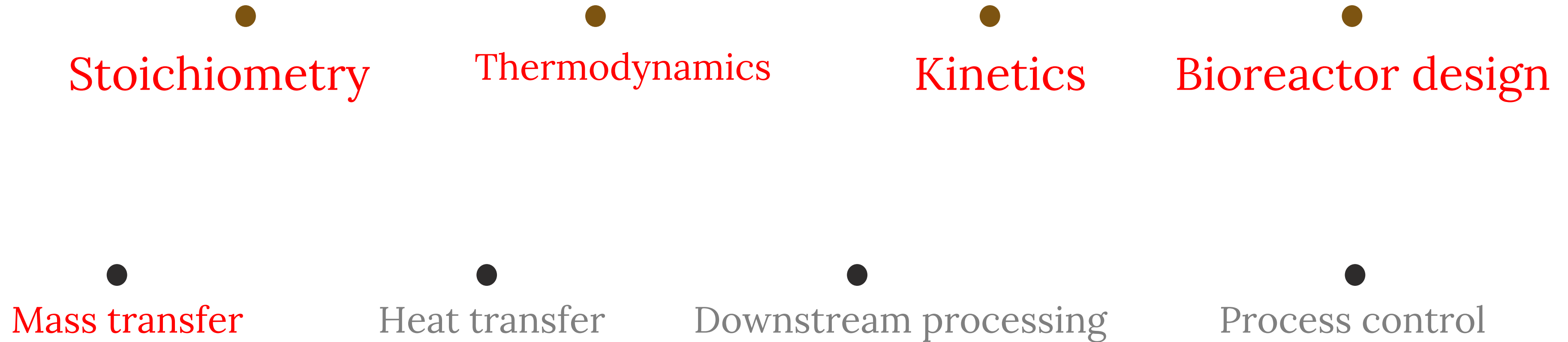
How much?

SEPARATIONS

How pure?

THE ROAD AHEAD

Major topics



THE FIELD IS MOVING FAST

Future directions



AI / ML
in bioprocessing



Automation



Continuous
bioprocessing



Cell-free
systems



Synthetic
biology

WHAT TO READ

Textbooks & references

Primary

Bailey & Ollis

BIOCHEMICAL ENGINEERING
FUNDAMENTALS

Core theory & problem sets.

Secondary

Nielsen, Villadsen & Lidén

BIOREACTION ENGINEERING PRINCIPLES

Rigorous stoichiometric & metabolic modelling.

Supplementary

Shuler & Kargi

BIOPROCESS ENGINEERING: BASIC CONCEPTS

Accessible worked examples & broad overview.

Grading and other details



-
- Exams (Mid sem and End sem) 40%
-
- Quizzes (3-4 quizzes) 20%
-
- Course project 40%
-

Office hours:

- Day: Monday & Tuesday
- Time: 12:30 – 1 PM
- Or by appointment

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